

MCQs :-

1. ~~A~~ B

2. C

3. A

4. D

5. A

6. D

7. ~~B~~ B

8. ~~B~~ C

9. B

10. ~~DC~~

11. D

12. A

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Section - B

Q: 2(i) ———
A: 2(i) ———

Elasticity:-

The ~~the~~ ability of the deforming body to return to its original shape and size when the deforming force is removed.

Examples:-

Spring etc.

Elastic limit:-

Most materials are elastic up to a certain limit known as elastic limit. Beyond the ~~the~~ elastic limit a material will not return to its original dimensions when the deforming force is removed.

Inelastic examples are:-

Clay, dough etc.

④

Q:2(ii)

A:2(ii)

[Statement]:

Within elastic limit the extension x is directly proportional to the restoring force. i.e. :-

$$F_{\text{res}} = -kx$$

$$k = \frac{-F_{\text{res}}}{x}$$

Spring constant :-

The ratio of the restoring force to the extension. It represents the stiffness of the spring.

SI unit = Nm^{-1}

Q:2(iii)

A:2(iii)

Data :-

$$m = 60 \text{ kg}$$

$$x = 8 \text{ cm} \rightarrow 0.08 \text{ m}$$

$$g = 10 \text{ m/s}^2$$

$$F = w = mg = 60 \times 10 = 600$$

$$k = \frac{600 \text{ N}}{0.08 \text{ m}} = 7500 \text{ N/m}$$

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Q:2(iv)(or)A:2(iv)(or)

Data:-

$$m = 50 \text{ Kg}$$

$$A = 0.02 \text{ m}^2$$

$$g = 9.8 \text{ m/s}^2$$

$$F = W = mg = 50 \times 9.8 = 490 \text{ N}$$

$$P = \frac{F}{A}$$

$$\Rightarrow \frac{490 \text{ N}}{0.02 \text{ m}^2} \Rightarrow 24500 \text{ Pa}$$

$$\Rightarrow 24.5 \text{ KPa}$$

Q:2 (ix)(or)A:2(v)(or)

Drinking by straw:-

The drinking through straw is possible by lowering the pressure in the mouth below atmospheric pressure. The action of sucking increases the volume of lungs, thereby reducing the air pressure in the lungs and the mouth. The atmospheric

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pressure acting on the surface of the liquid will be greater ~~than~~ then be greater the pressure in the mouth, thus forcing the liquid to rise up the straw into the mouth.

Drawing liquid by syringe :-

Pulling the piston back decreases pressure inside the cylinder. Atmospheric pressure acting on the liquid surface outside then outside then drives the liquid into the syringe through the needles.

$$\begin{array}{c} \text{--- } Q : 2(v_i) \text{ ---} \\ \text{--- } A : 2(v_i) \text{ ---} \end{array}$$

Construction :-

A long glass tube (almost 80 cm) closed at one end is filled with mercury and inverted into a container (trough) of mercury. The closed end creates a vacuum above the mercury column.

Working :-

The mercury in the tube falls until the weight of the mercury column is exactly balanced by the atmospheric pressure pushing down on the mercury in the trough. It works on the principle of hydrostatic equilibrium.

Height represents :-

The height h of the mercury column directly represents atmospheric pressure. At sea level $h = 760 \text{ mm Hg} = 76 \text{ cm} = 1 \text{ atm} = 101.325 \text{ kPa}$.

Q: 2 (vii) (or)
A: 2 (vii) (or)

Data :-

$$\begin{aligned} h &= 500 \text{ m} \\ \rho &= 1025 \text{ Kg/m}^3 \\ g &= 9.8 \text{ m/s}^2 \\ P &= \rho gh \\ P &= 1025 \times 9.8 \times 500 \\ P &= 5022500 \rightarrow 5.0225 \times 10^6 \\ &= 5.02 \text{ MPa} \end{aligned}$$

Q:2(viii)(or)
A:2(vii)(or)

Applications of manometer :-

- 1) It is used to measure vacuum.
- 2) It is ~~also~~ also used to measure calibrations.
- 3) It is used to measure leak testing.
- 4) It is used to measure the flow of the fluid.

Q:2(ix)
A:2(ix)

Manometer	Barometer
1) It is a device that is used to measure the pressure of the fluid but that of liquid concerning the earth's atmospheric pressure.	It is a device that is used to measure fluid pressure but that of air as it can differ with distance with it's below or above sea level.
2) It comes in different forms.	It comes only in one basic design.

for all its types.

These are filled with mercury or any heavy liquid material but in some cases, they can be filled with a liquid lighter liquid material.

Q: 2 (n) (or)

A: 2 (n) (or)

A) Brake Pedal:-

When we press the brake pedal, it exerts force on the master cylinder's piston.

B) Master cylinder:-

The force on the piston increases the pressure in the brake fluid.

C) Brake Lines:-

The increased pressure is transmitted equally through the brake fluid in the brake lines to all wheels.

Q) Brake calipers:-

At each wheel, the pressure acts on brake calipers or wheel cylinders, applying force to brake pads.

$$\begin{array}{c} \text{Q: } 2(x_i)(or) \\ \text{A: } 2(x_i)(or) \end{array}$$

Drivers wear special suits:-

At great depth, liquid pressure $P = \rho gh$ is enormous. Drivers need suits that can withstand this high inward pressure on all body surfaces. Without protection, the pressure would crush the driver's body, rupture blood vessels, and collapse the lungs.

At 8.5 km depth:-

$$\begin{aligned} P = \rho gh &= 1000 \times 9.8 \times 8500 = 8.33 \times 10^7 \text{ Pa} \\ &= 83.3 \text{ MPa} \end{aligned}$$

①

Section - C

Q.3 (i)
A.3 (i)

Elasticity:-

The ability of a deformed body to return to its original shape or size when the deforming forces on it is removed is called elasticity.

Elastic limit:-

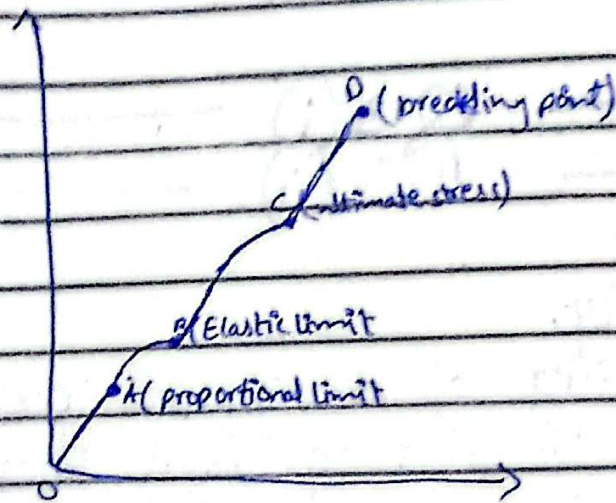
Most materials are elastic up to a certain limit is known as Elastic limit. Beyond the elastic limit, a material will not return to its original dimension when the deforming force is removed.

Hooke's law:-

Within elastic limits the extension (or compression) is directly proportional to the restoring force. F i.e.

$$F_{\text{rest}} \propto x \quad \text{or} \quad F_{\text{rest}} = -Kx$$

②



$$\frac{Q}{A} = \frac{3}{3} \quad \text{(i) (or)} \quad \frac{Q}{A} = \frac{3}{3} \quad \text{(ii) (or)}$$

Atmospheric pressure :-

The pressure that atmospheric particles exert on the surface of the earth and all over the surface of objects on the earth is called atmospheric pressure. At the sea level, the pressure of the atmosphere on average is $1.013 \times 10^5 \text{ Pa}$.

Liquid Barometer :-

A liquid barometer is a device that measures atmospheric pressure using the principles of principles and the behaviour of liquid.

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The most common type of the liquid barometer is the mercury barometer.

Mercury barometer :-

The liquid barometer works on the principle of hydrostatic equilibrium, which states that the pressure at any point in a fluid at rest is the same at all depths.

At sea level, the atmosphere will push down mercury in the tube and make it rise up in a tube to a height of approximately 760 millimeters (mmHg) or 29.92 inches $\approx$ 76 cm.

Variation with altitudes :-

As altitudes increase, the weight of air column above decreases, so atmospheric pressure decreases. $1 \text{ atm} = 760 \text{ mmHg}$ at sea level; much less at high altitudes.

(u)

Weather indication :-

Falling barometer reading indicates approaching low pressure (storms, rain). Rising reading indicates high pressure (clear, fair weather). Weather maps use isobars (lines of equal pressure) in millibars (1 mbar = 100 Pa).

Q:3 (iii)
A:3 (iii)

A cylindrical liquid column includes :-

- * height = h
- * Base = A
- * Liquid density = ρ

Derivation :-

$$F = W = mg$$

$$\Rightarrow \rho = \frac{m}{V} \quad \left(V = \frac{Ah}{A} \right)$$

$$\Rightarrow \rho = \frac{F}{Ahg}$$

(5)

$$\frac{P}{\rho h g}$$

$$P = \rho h g$$

Liquids pressure depends only on density ρ and depth h - not on the shape or total volume of the container.

Drill-hole activity :-

water shoots fastest and ~~further~~ furthest from the lowest hole because pressure is greatest at greater depth.

(2)

Data :-

$$h = 8.5 \text{ km}$$

$$\rho = 1000 \text{ kg/m}^3$$

$$g = 9.8 \text{ m/s}^2$$

converting km to m.

$$8.5 \times 1000 = 8500 \text{ m}$$

$$P = \rho h g$$

$$= 1000 \times 8500 \times 9.8$$

$$= 8330000$$

$$= 83.3$$

⑥

Q: 3 (iv)

A: 3 (iv)

Pascal Principle:-

An external pressure applied to an enclosed fluid is transmitted unchanged to every point within the fluid.

hydraulic lift Derivation:-

Piston ① = A_1

Piston ② = A_2

Pressure of Piston 1 = $P_1 = \frac{F_1}{A_1}$

" " " 2 = $P_2 = \frac{F_2}{A_2}$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$P_1 = P_2$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_2 = \frac{A_2}{A_1} \times F_1$$

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② Data :-

$$A_1 = 0.002 \text{ m}^2$$

$$A_2 = 0.9 \text{ m}^2$$

$$m = 1800 \text{ Kg}$$

$$g = 9.8 \text{ m/s}^2$$

$$F_1 = W = mg$$

$$1800 \times 9.8 = 17640 \text{ N}$$

$$F_2 = \frac{0.9 \text{ m}^2}{0.002 \text{ m}^2} \times 17640$$

$$= \frac{0.002 \text{ m}^2}{0.9 \text{ m}^2} \times 17640$$

$$= 39.2 \text{ N}$$